1. Game Overview

Title: *Project BiFractal*

Genre: 2D Puzzle Platformer

Target Audience: Teenagers and young adults (ages 13–30

Platform: PC (Unity Engine – 2D)

Project BiFractal is a 2D puzzle-platformer inspired by physics-based problem-solving games of the likes of Valve's *Portal*. Players take control of an inmate forced to test experimental technologies in a controlled lab environment. By solving increasingly complex puzzle rooms, players progress through the game while uncovering subtle narrative elements.

2. Game Objective

The primary objective of the game is to complete puzzle rooms by using standard 2D platformer movement and gadgets such as a gravity manipulation device.

- **Success Condition:** Reaching the exit point of each room
- **Failure Condition:** Contact with hazards (spikes, lasers, crushing mechanisms), resulting in instant death and restart from the last checkpoint
- **Progression System:**
    - Linear level-based progression
    - Increasing difficulty
    - Gradual unlocking of new mechanics (i.e. gravity manipulation)
- **No scoring system**, but performance may be measured by time (*optional feature*)

The gameplay is designed around experimentation and puzzle-solving. It encourages players to come up with creative solutions with the tools at their disposal.

3. MDA Framework

3.1 Mechanics

- **Player Actions:**
  - Move left and right
  - Jump
  - Use gadgets (i.e. gravity manipulation)
- **Core System:**
  - Gravity manipulation (flip gravity to all four cardinal directions)
- **Game Rules:**
  - Player dies instantly upon touching hazards
  - Player must reach the exit to progress
- **Resources:**
  - No health system (instant death)
  - Unlimited lives
- **Constraints:**
  - Limited movement space
  - Timing and precision required
- **Checkpoints:**
  - Checkpoints are placed at the start of each puzzle room

3.2 Dynamics

- Players experiment with gravity to navigate obstacles
- Thinking creatively becomes central to progression
- Increasing complexity forces players to:
    - Plan ahead
    - Adapt quickly
    - Combine mechanics creatively

Emergent Gameplay:

- Players may discover unintended solutions
- Speedrunning strategies may emerge
- Risk-taking vs. safe play decisions

Game Response:

- Immediate feedback on failure (death then reset)
- Smooth transitions between attempts encourage persistence and experimentation

3.3 Aesthetics

The game aims to evoke:

- **Challenge:** Difficult puzzles requiring precision
- **Curiosity:** Discovering how each new mechanic works
- **Tension:** Avoiding deadly hazards
- **Satisfaction:** Solving complex puzzles

Visual Style:

- Minimalist, laboratory-themed environments
- Cold, sterile color palette (whites, grays and blues)

Audio Design:

- Ambient, mechanical background sounds
- Subtle music to enhance tension
- Sharp sound effects for hazards and death to convey danger and failure

4. Formal Elements

Player Interaction

- Single-player experience
- Direct control of character movement and gadget usage

Rules

- Avoid hazards
- Solve puzzles using available tools
- Reach the exit to progress

Procedures

1. Enter a room
2. Analyze puzzle layout
3. Maneuver around and use gadgets
4. Complete the room

Resources

- Player abilities (movement and gadgets)
- Instant death, but unlimited lives

Boundaries

- Confined 2D puzzle rooms
- No open-world exploration

Outcomes

- Success: Completing the puzzle room
- Failure: Touching a hazard and respawning

5. Dramatic Elements

Premise

The player is an inmate forced to repay their debt to society by acting as a test subject for experimental technologies in a controlled lab.

Character

- **The Inmate (Player Character):**
  - Silent protagonist
  - Identity left ambiguous to increase immersion

Story Progression

- Each level represents a new test chamber
- Between levels, the player is cryogenically frozen and transported to the next one
- Environmental storytelling hints at:
  - The nature of the experiments
  - The organization behind the tests
  - Disapproval of the public

6. Gameplay Evolution

Advanced Strategies:

- Timing gravity shifts mid-air
- Memorizing hazard patterns
- Optimizing routes

Emergent Behaviors:

- Creative puzzle-solving beyond intended solutions
- Experimentation with physics interactions
- Mastery-based gameplay (fewer deaths over time)

Varied Outcomes:

- Different players solve puzzles differently
- Potential for alternative routes or unintended shortcuts

7. Elemental Tetrad

Mechanics

- Core gameplay revolves around movement and gravity manipulation

Story

- Minimalist but meaningful narrative integrated into gameplay progression through environmental storytelling

Aesthetics

- Clean, sterile visuals reinforce the laboratory setting
- Audio enhances immersion and tension

Technology

- Unity engine enables:
    - Physics simulation
    - Smooth 2D gameplay
    - Scalable level design

Interconnection:
All four elements work together to create a cohesive experience where:

- Mechanics support puzzle-solving
- Aesthetics reinforce the lab environment and the stressful situation
- Story adds deeper meaning without interrupting gameplay
- Technology enables seamless interaction

8. Conclusion

Project BiFractal is a physics-based puzzle platformer that combines elaborate puzzles with a subtle narrative. By applying the MDA framework and formal design principles, the game delivers a cohesive and engaging experience centered around experimentation, failure and mastery of the game's mechanics.